

Insights Into Penguin Characteristics Using Data Visualization

Shubham Prakash

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```
library(tidyr)
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ purrr      1.0.2
## ✓ forcats    1.0.0      ✓ readr      2.1.5
## ✓ ggplot2    3.5.1      ✓ stringr    1.5.1
## ✓ lubridate  1.9.3      ✓ tibble     3.2.1
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(palmerpenguins)
library(ggthemes)
library(ggcorrplot)
library(gganimate)
library(gifski)
library(transformr)
```

```
data("penguins")
df<- penguins
class(df)
```

```
## [1] "tbl_df"      "tbl"        "data.frame"
```

```
str(df)
```

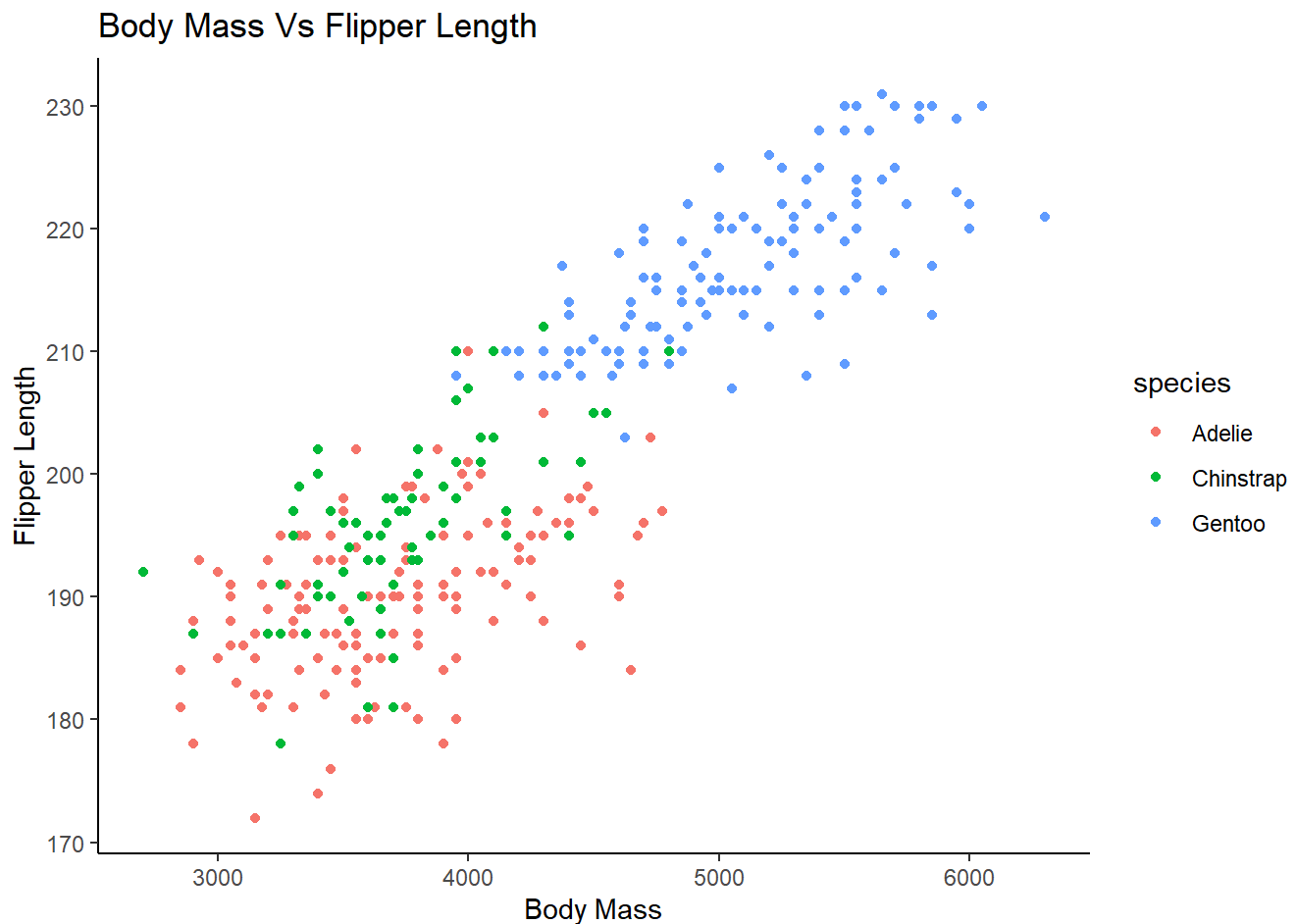
```
## tibble [344 × 8] (S3: tbl_df/tbl/data.frame)
## $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3 ...
## $ bill_length_mm: num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
## $ bill_depth_mm: num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
## $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
## $ body_mass_g   : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
## $ sex          : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
## $ year         : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

```
head(df)
```

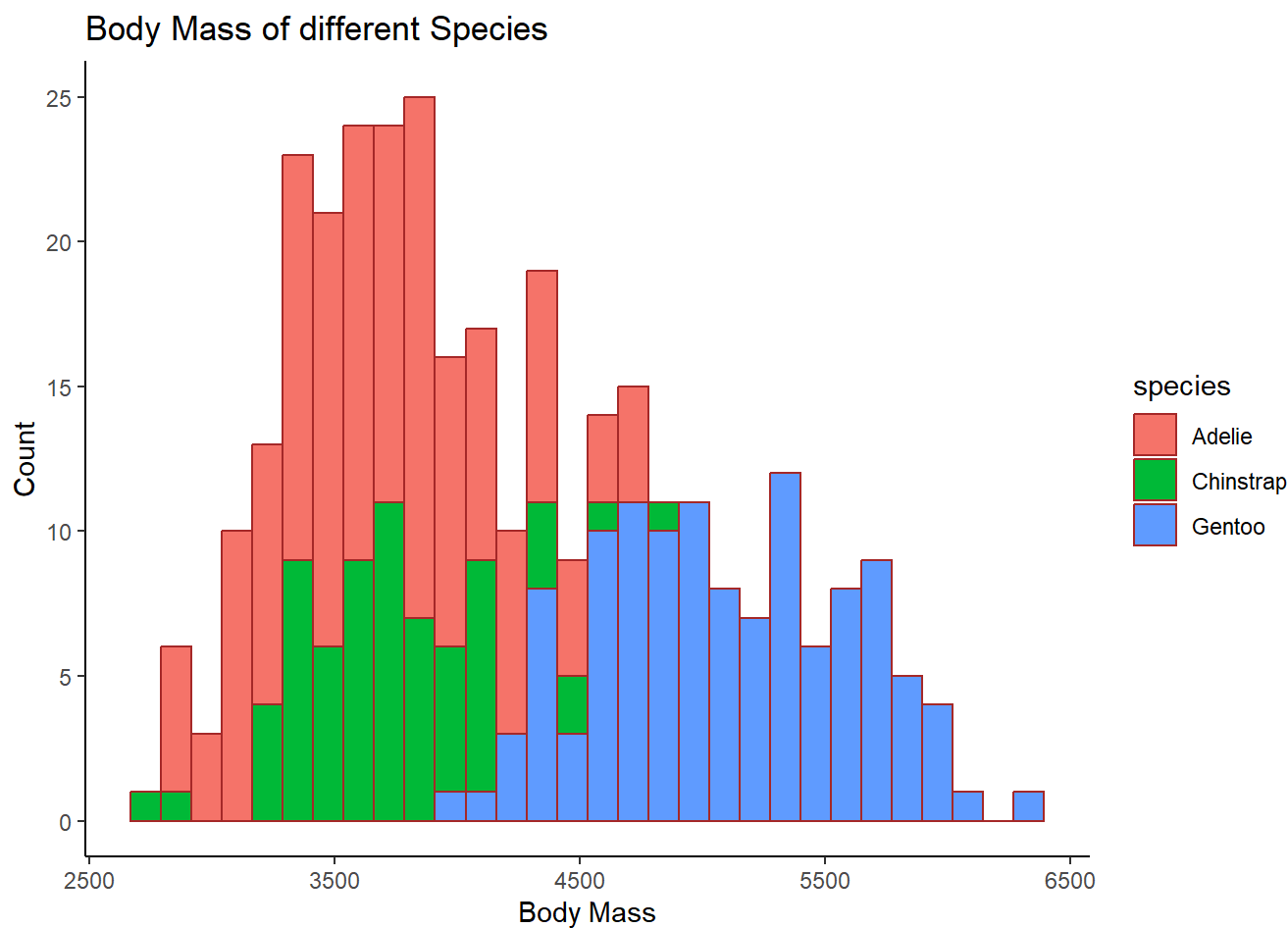
```
## # A tibble: 6 × 8
##   species island   bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>         <dbl>         <dbl>         <int>         <int>
## 1 Adelie  Torgersen         39.1          18.7          181          3750
## 2 Adelie  Torgersen         39.5          17.4          186          3800
## 3 Adelie  Torgersen         40.3          18           195          3250
## 4 Adelie  Torgersen         NA           NA           NA           NA
## 5 Adelie  Torgersen         36.7          19.3          193          3450
## 6 Adelie  Torgersen         39.3          20.6          190          3650
## # i 2 more variables: sex <fct>, year <int>
```

```
pp<-na.omit(df)
```

```
# Scatter Plots..
ggplot(pp, aes(body_mass_g, flipper_length_mm, col= species))+
  geom_point()+
  labs(x= "Body Mass", y = "Flipper Length" , title = "Body Mass Vs Flipper Length")+
  theme(plot.title = element_text(hjust = 0.5))+ theme_classic()
```



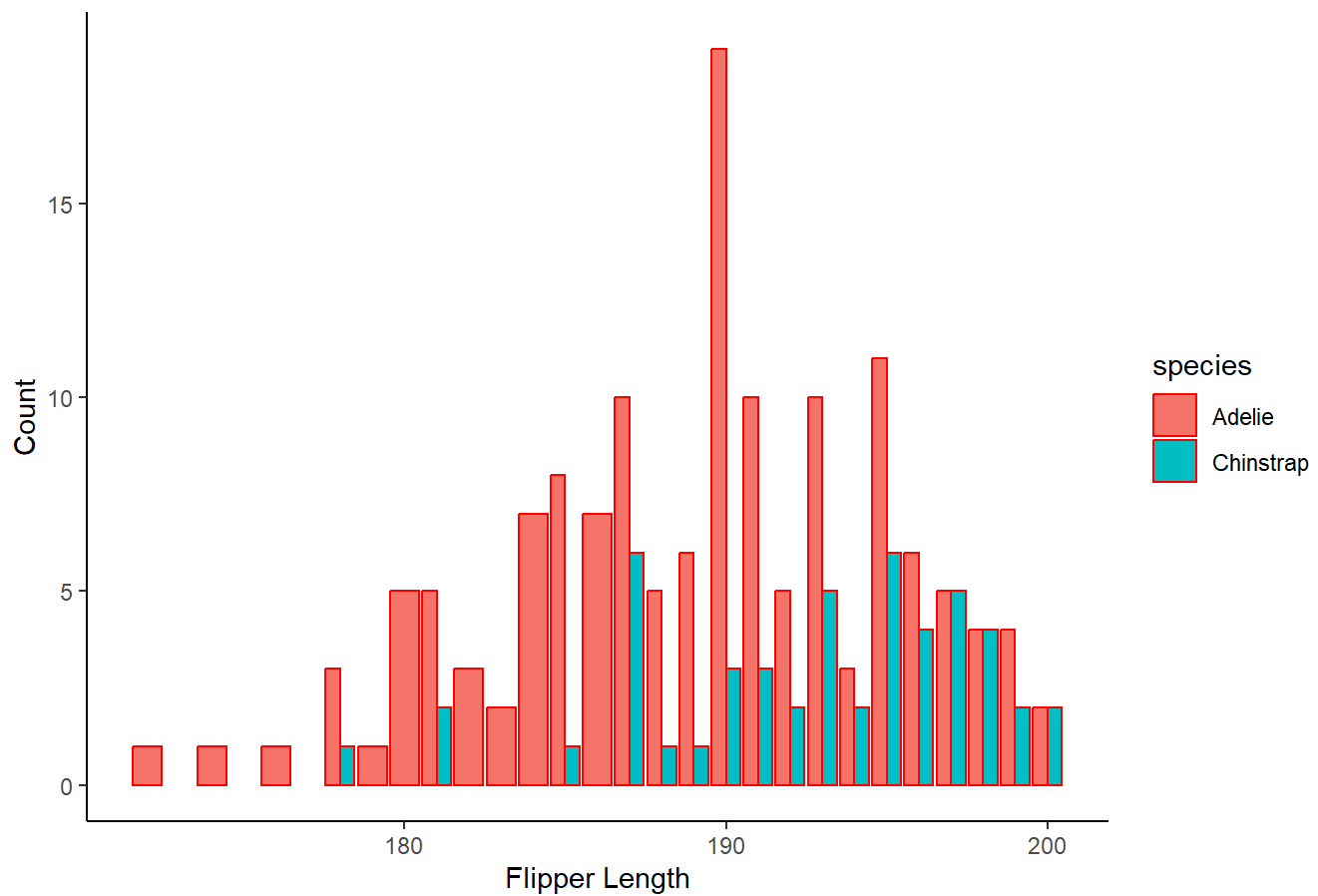
```
# Histogram..
ggplot(pp, aes(body_mass_g, fill = species))+
  geom_histogram(col = 'brown', bins = 30)+
  labs(x= "Body Mass", y = "Count" , title = "Body Mass of different Species")+
  theme(plot.title = element_text(hjust = 0.5))+ theme_classic()
```



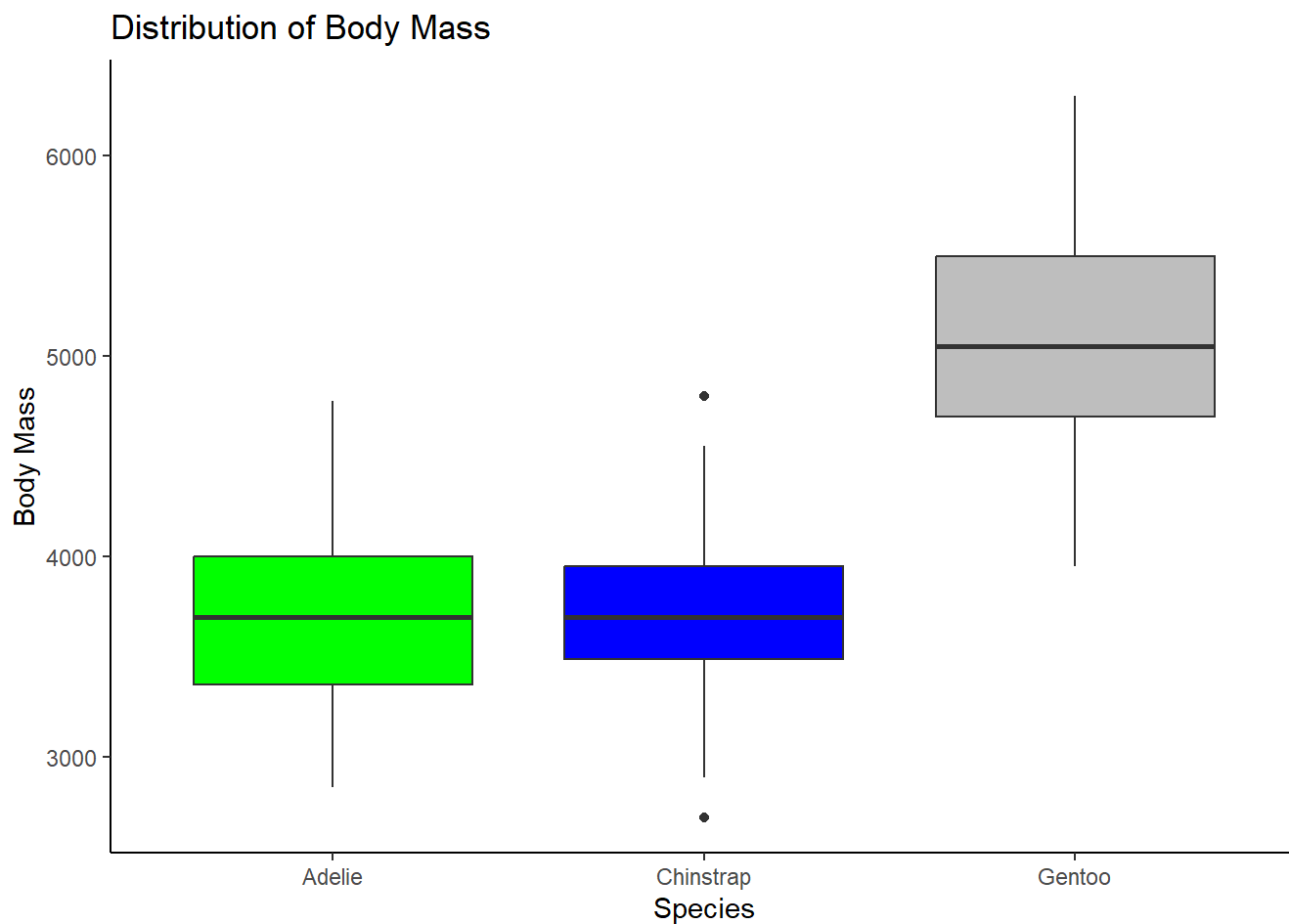
```
ff<-df %>% filter(flipper_length_mm <= 200)

# Bar Plot..
ggplot(ff, aes(flipper_length_mm, fill = species))+
  geom_bar(col = 'red', position = "dodge")+
  labs(x= "Flipper Length", y = "Count" , title = "Flipper Length of different Species")+
  theme(plot.title = element_text(hjust = 0.5))+ theme_classic()
```

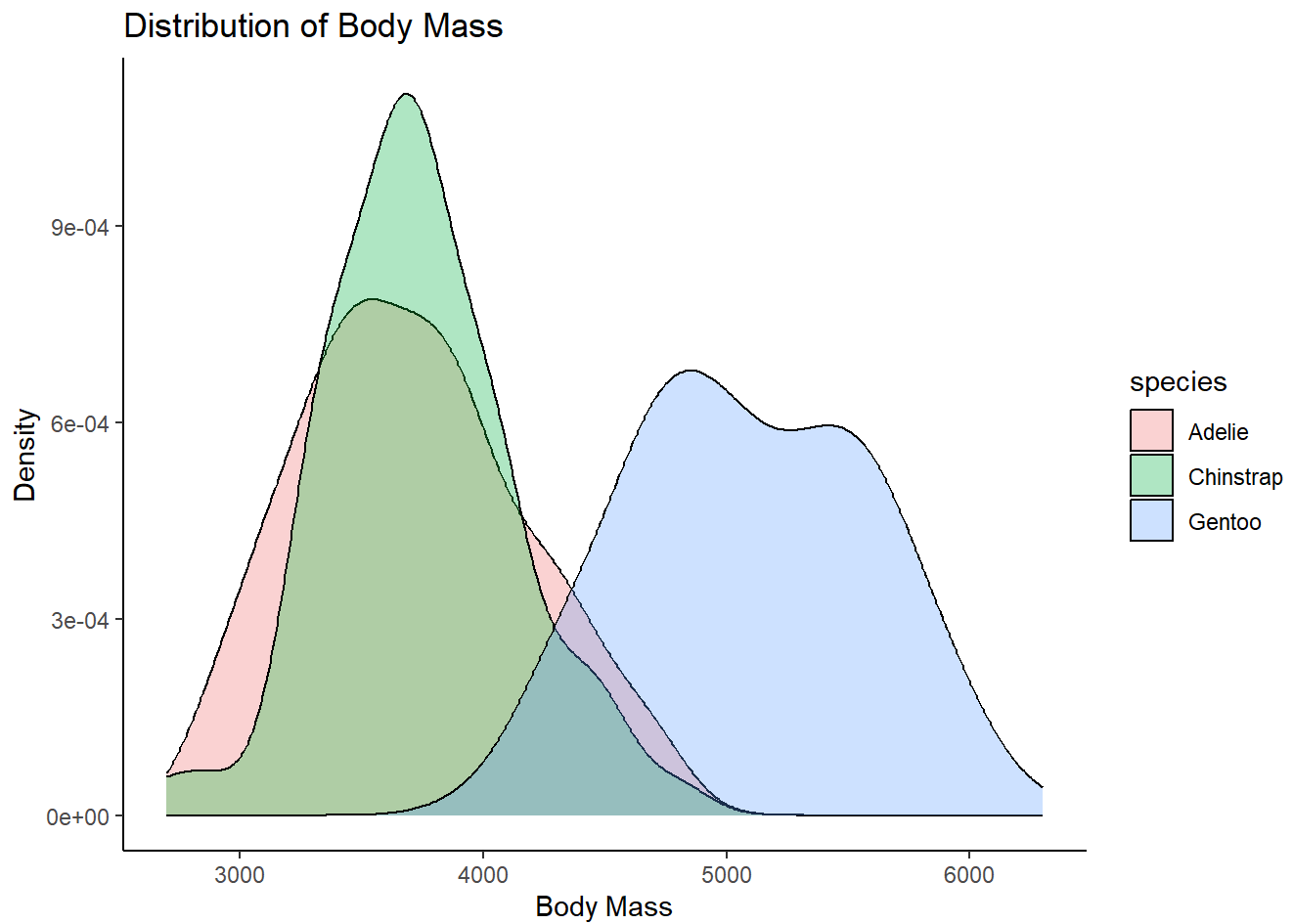
Flipper Length of different Species



```
# Box Plot..
ggplot(pp, aes(species,body_mass_g))+
  geom_boxplot(fill = c('green','blue','grey'))+
  labs(x= "Species", y = "Body Mass" , title = "Distribution of Body Mass")+
  theme(plot.title = element_text(hjust = 0.5))+ theme_classic()
```



```
# Density Plot..  
ggplot(pp, aes(body_mass_g, fill = species ))+  
  geom_density(alpha = 0.3)+  
  labs(x= "Body Mass", y = "Density" , title = "Distribution of Body Mass")+  
  theme(plot.title = element_text(hjust = 0.5))+ theme_classic()
```



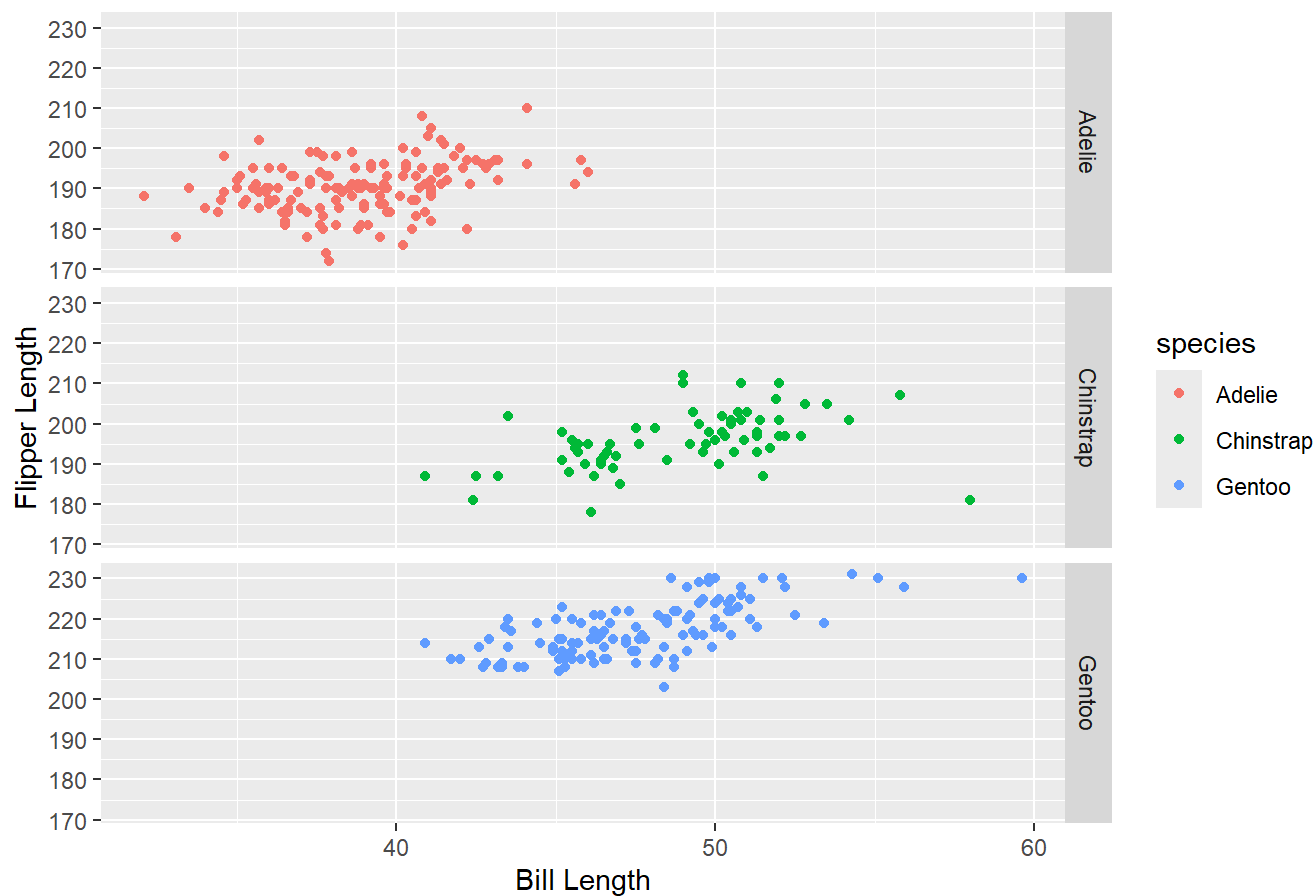
```
# Facet Grids..  
ggplot(pp, aes(body_mass_g, flipper_length_mm))+  
  geom_point(aes(col = species))+ facet_grid(~species) +  
  labs(x= "Body Mass", y = "Flipper Length" , title = "Flipper Length Vs Body Mass among Species") +  
  theme(plot.title = element_text(hjust = 0.5))
```

Flipper Length Vs Body Mass among Species



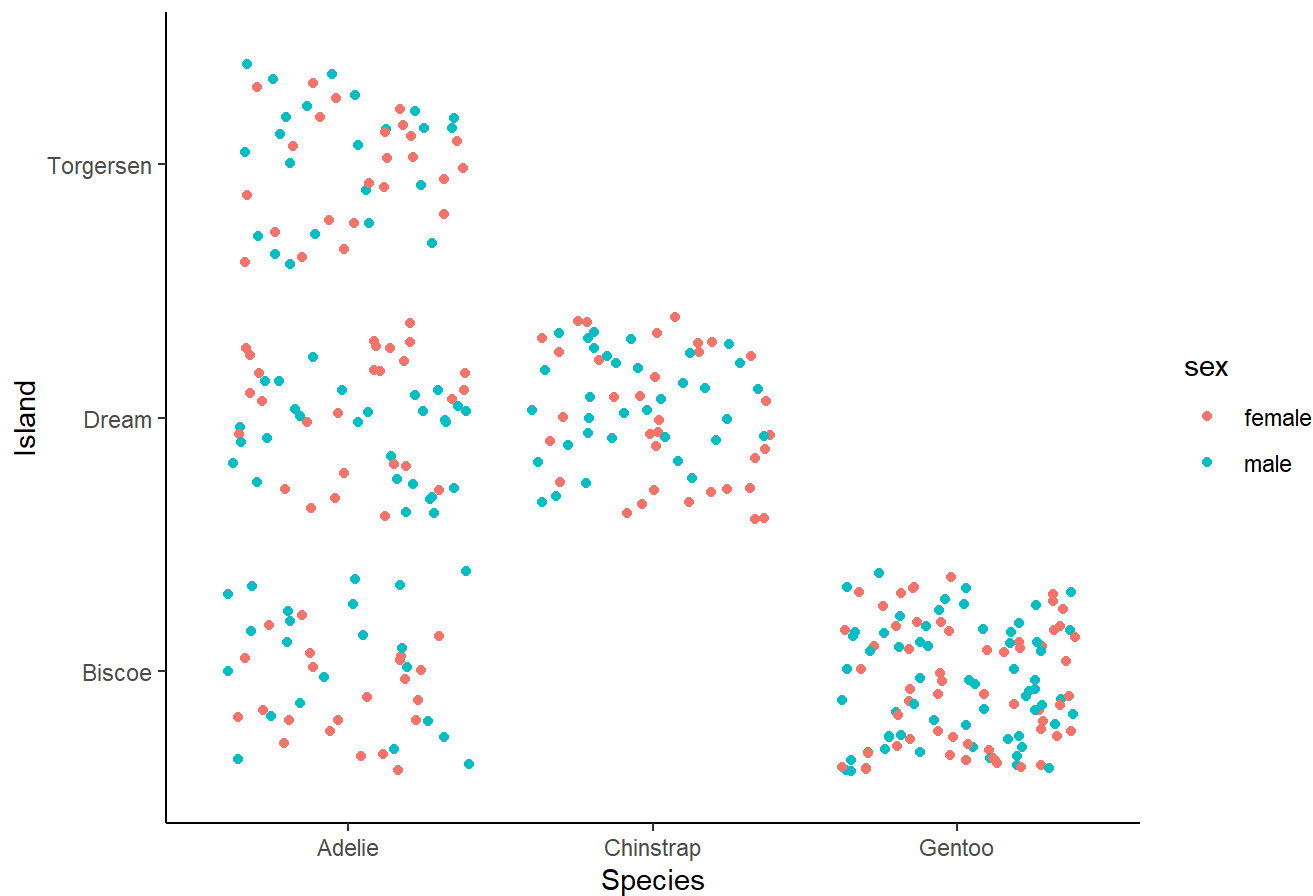
```
ggplot(pp, aes(bill_length_mm, flipper_length_mm))+  
  geom_point(aes(col = species))+ facet_grid(species~.) +  
  labs(x= "Bill Length", y = "Flipper Length" , title = "Flipper Length Vs Bill Length among Species")+  
  theme(plot.title = element_text(hjust = 0.5))
```

Flipper Length Vs Bill Length among Species



```
# Geom Jitter..
ggplot(pp, aes(species, island , colour = sex))+
  geom_jitter()+ labs(x= "Species", y = "Island" , title = "Distribution by Species, Island and
Sex")+
  theme(plot.title = element_text(hjust = 0.5))+ theme_classic()
```


Distribution by Species, Island and Sex



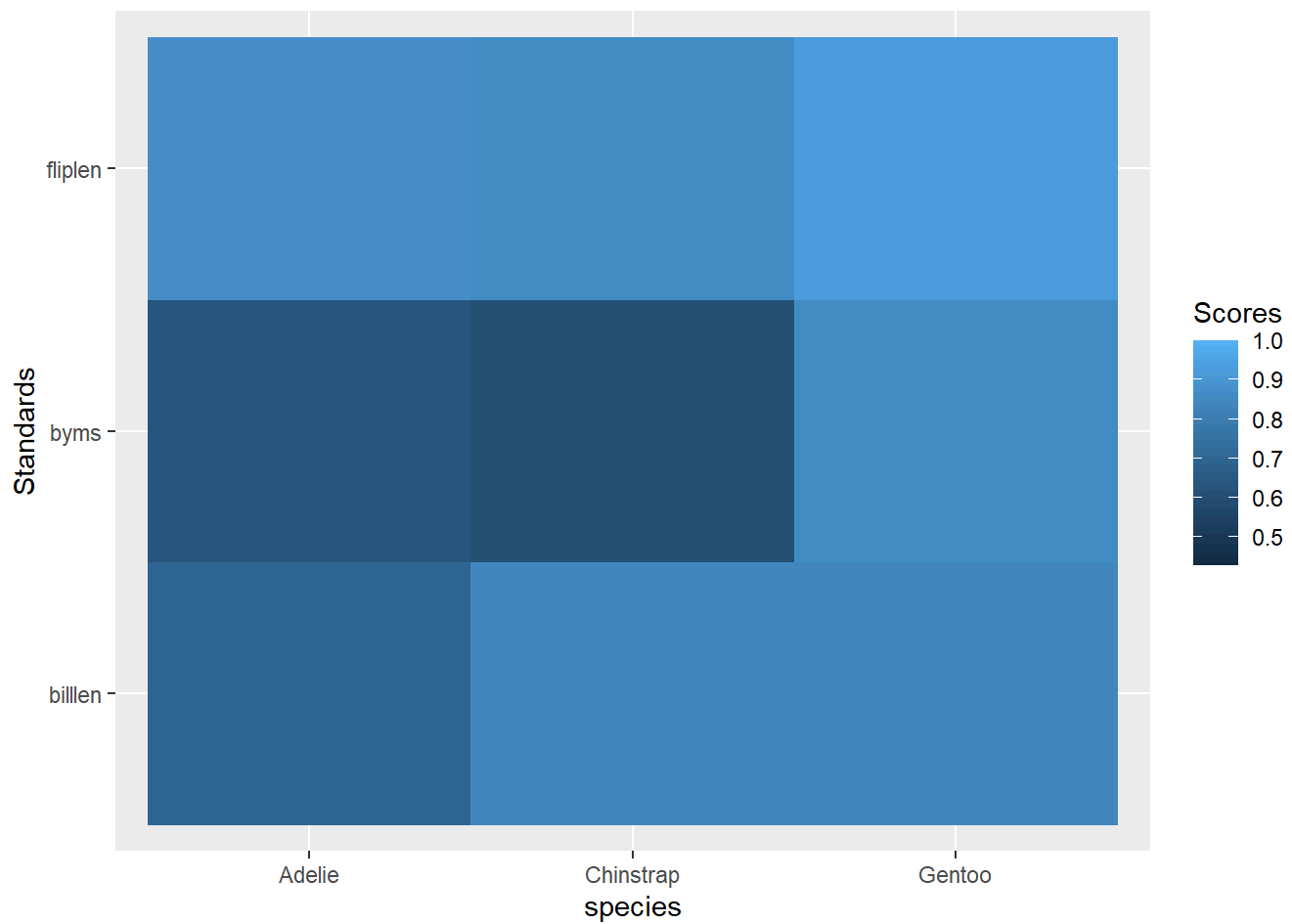
```
# Heatmaps..
pp_sub<- pp %>%
  select(species, bill_length_mm, flipper_length_mm, body_mass_g)

pp_sub$byms<- pp_sub$body_mass_g/max(pp_sub$body_mass_g)
pp_sub$fliplen<- pp_sub$flipper_length_mm/max(pp_sub$flipper_length_mm)
pp_sub$billllen<- pp_sub$bill_length_mm/max(pp_sub$bill_length_mm)

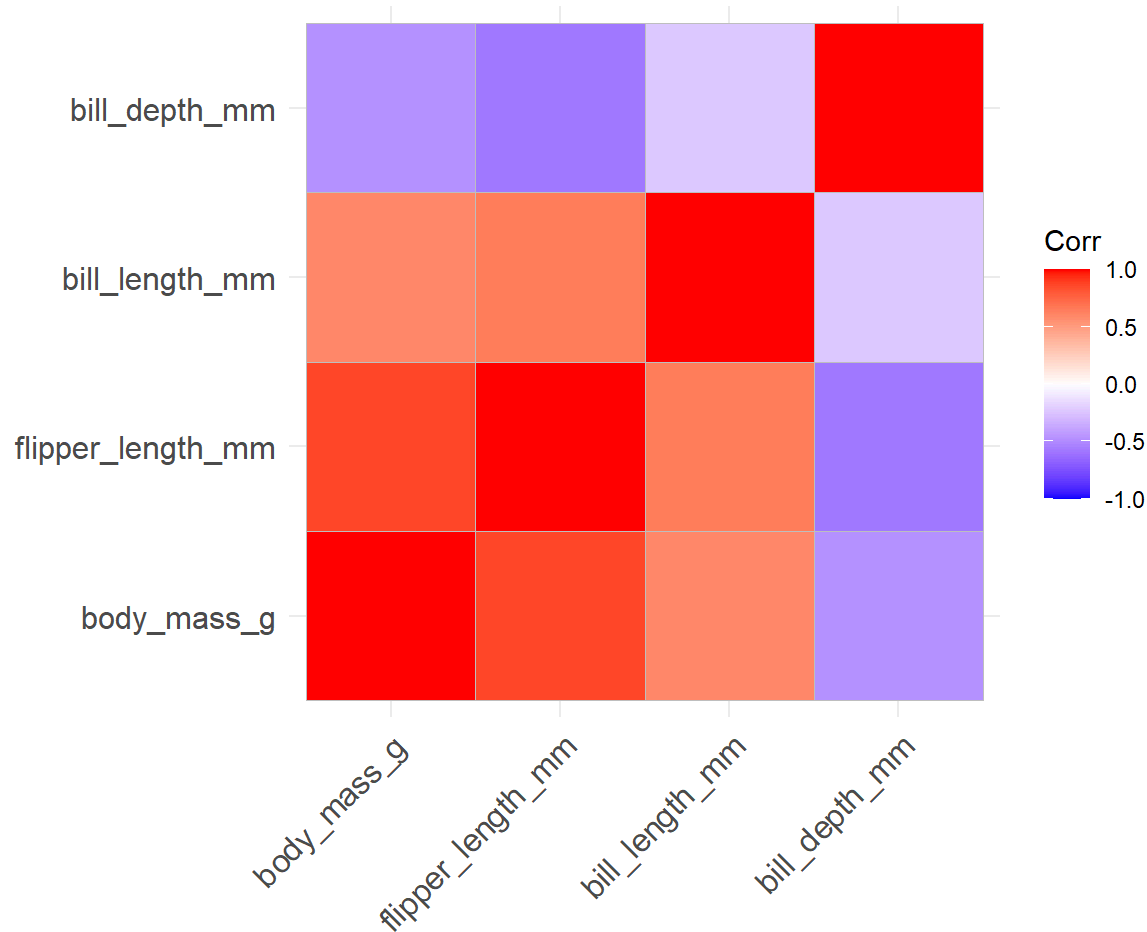
pp_sub2<- pp_sub %>%
  select(c("species", "byms", "fliplen", "billllen"))

pp_sub3<- pivot_longer(pp_sub2,
  c("byms", "fliplen", "billllen"),
  names_to = "Standards", values_to = "Scores")

ggplot(pp_sub3, aes(species, Standards, fill= Scores))+
  geom_tile()
```

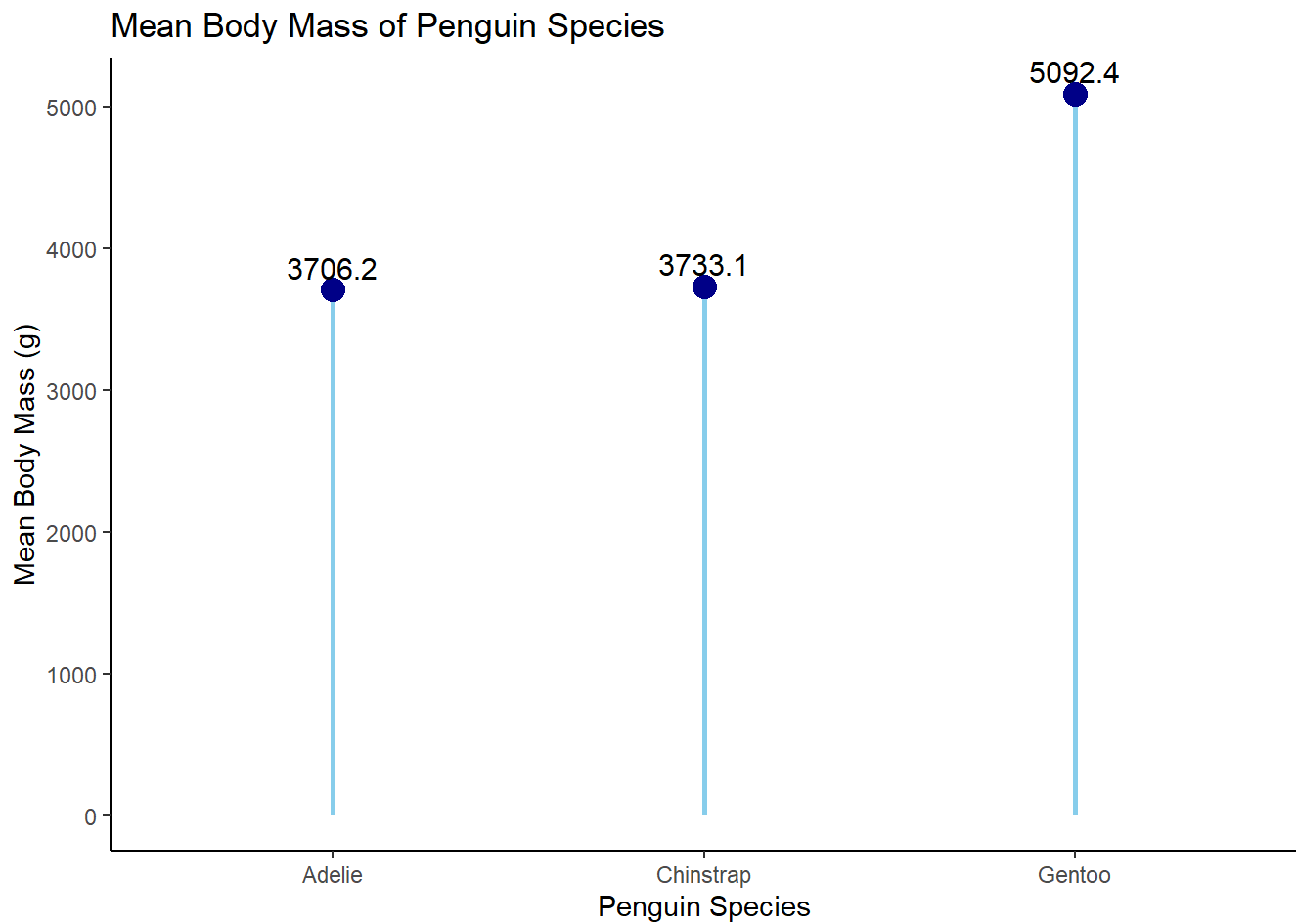


```
# Correlation Plots..  
pp_2<- pp %>%  
  select(body_mass_g,flipper_length_mm,bill_length_mm, bill_depth_mm)  
a<- cor(pp_2, use = "complete.obs")  
ggcorrplot(a)
```

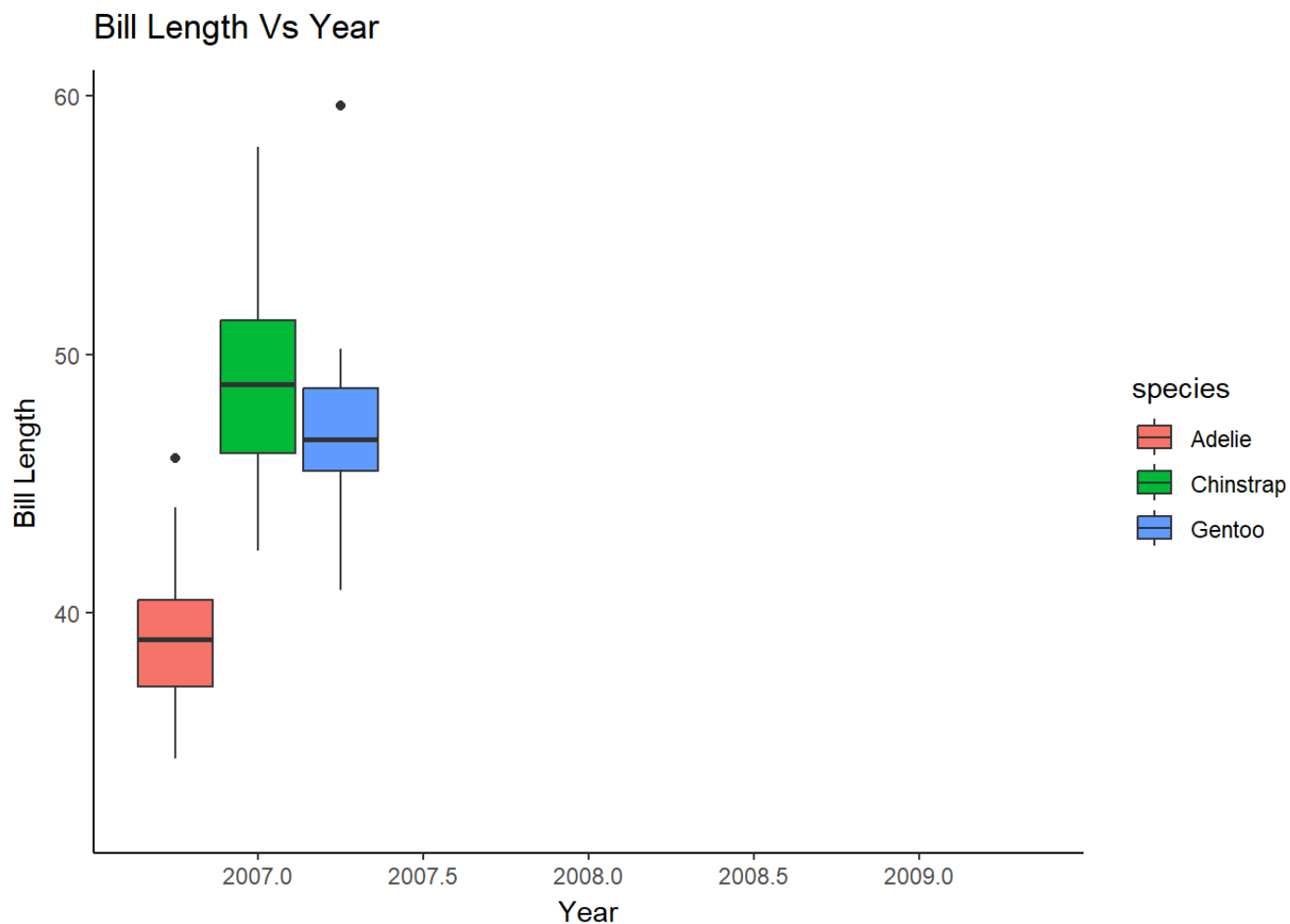


```
# Lollypop Plot..
penguins_summary <- pp %>%
  group_by(species) %>%
  summarise(mean_body_mass = mean(body_mass_g))
ggplot(penguins_summary, aes(x = species, y = mean_body_mass)) +
  geom_segment(aes(x = species, xend = species, y = 0, yend = mean_body_mass),
    color = "skyblue", size = 1) +
  geom_point(color = "darkblue", size = 4) +
  geom_text(aes(label = round(mean_body_mass, 1)),
    vjust = -0.5, color = "black", size = 4) +
  labs(title = "Mean Body Mass of Penguin Species", x = "Penguin Species",
    y = "Mean Body Mass (g)") + theme_classic()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



```
# Animated Graphs..  
ggplot(pp, aes(year, bill_length_mm, fill = species))+  
  geom_boxplot()+transition_time(year)+labs(x= "Year", y = "Bill Length" ,  
                                             title = "Bill Length Vs Year")+  
  theme(plot.title = element_text(hjust = 0.5))+ theme_classic()
```



```
ggplot(pp, aes( body_mass_g, fill = species))+
  geom_histogram(col= 'black', bins = 30 )+
  transition_states(species)+ labs(x= "Body Mass", y = "Count" ,
                                title = "Body Mass Vs Count")+
  theme(plot.title = element_text(hjust = 0.5))+ theme_classic()
```

